

Do Attention Heads in BERT Track Syntactic Dependencies?

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BACKGROUND:

- Is there some structure in the attention weights of BERT models that map to our understanding of dependency relations?

DEPENDENCY EXTRACTION METHODS

1. Maximum attention weights (MAX)

Assign a relation (w_i, w_j) between the two words w_i and w_j if $j = \operatorname{argmax}_i W[i]$ for each row i of the attention matrix W . This is to analyze whether there are specialized heads that track dependency relations.

2. Maximum Spanning Tree (MST)

Use gold root of dependency tree as the starting node and apply the Chu-Liu-Edmonds algorithm on the attention weight matrix W to compute the maximum spanning tree. The resulting tree is a valid directed dependency tree. This is to investigate if the BERT attention heads are forming complete, syntactically informative parse trees.

Models

- BERT
- BERT fine-tuned on CoLA
- BERT fine-tuned on MNLI
- RoBERTa

BERT and RoBERTa have some specialized attention heads that track specific dependency types, but they do not have generalist heads that can do holistic parsing

RESULTS

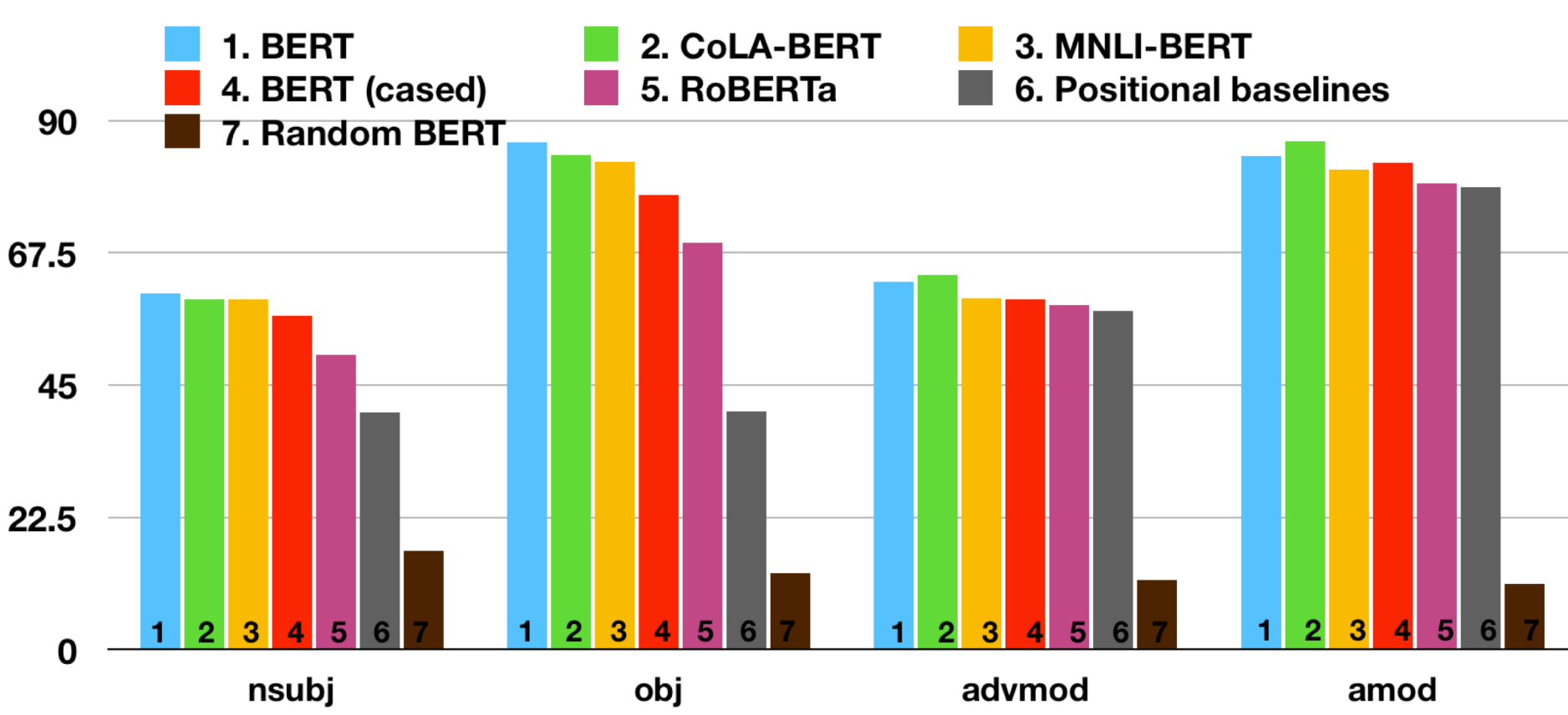


Figure 1. Undirected dependency accuracies by type based on our maximum attention weight method.

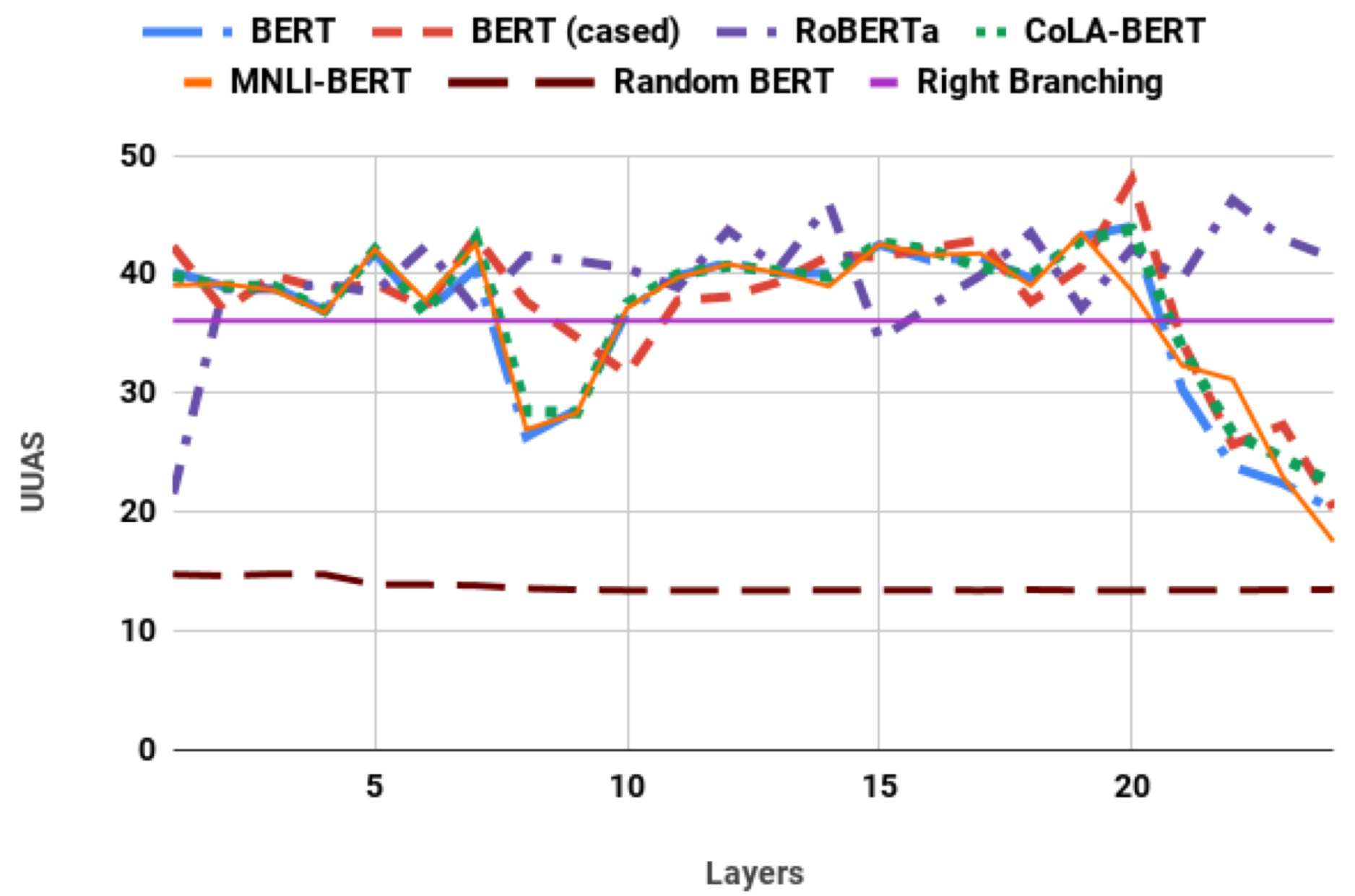


Figure 2. Maximum UUAS across layers of dependency trees extracted based on the maximum spanning arborescence on PUD

- Vanilla BERT and CoLA-BERT generally outperform the MNLI-BERT, except advcl and csubj, which are long-distance dependencies

